Nanyang Technological University
School of Electrical & Electronic Engineering
IE2106 Engineering Mathematics I

Tutorial 12 PDE

discuss in class

1. For each differential equation (DE), state with reason

(i) the *order* of the DE

(ii) if the DE is *homogeneous*

(iii) if it is *ODE* or *PDE*

(iv) if it is *variant* or *invariant*

(v) if it is *linear* or *non-linear*

(a) $u_x + u_y = u_x u_y$

(b) $u_{xx} + u + xe^y = 0$

(c) $u_x + 3u_y + 3x\cos(xy) = -y^2 u_{xx}$

(d) $u_{xx} + u + x = 0$

(e) $u_{xx} + u + y = 0$

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2. Given the partial differential equation

$$u_{xy} = 2xe^y,$$

with the initial conditions

$$u(0, y) = y \qquad \text{and} \qquad u(x, 0) = x.$$

Solve for $u(x, y)$.

3. Given the PDE

$$z_{x\,y} = x^2 y$$

(a) Solve the PDE by direct integration.

(b) Find the particular solution for which

$$z(x, 0) = x^2 \qquad \text{and} \qquad z(1, y) = \cos y.$$

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4. Show that

$$u(x, t) = F(2x + 5t) + G(2x - 5t)$$

is the general solution of

$$4u_{tt} = 25u_{xx}.$$

Here, F and G are arbitrary functions.

Answers

2. $u(x, y) = x^2 e^y + y + x - x^2$

3. $u(x, y) = \frac{1}{6}x^3 y^2 + \cos y - \frac{1}{6}y^2 + x^2 - 1$